

- A circle passing through the point  $P(\alpha, \beta)$  in the first quadrant touches the two coordinate axes at the points  $A$  and  $B$ . The point  $P$  is above the line  $AB$ . The point  $Q$  on the line segment  $AB$  is the foot of perpendicular from  $P$  on  $AB$ . If  $PQ$  is equal to 11 units, then the value of  $\alpha\beta$  is -----  
[2023 (06 Apr Shift 1)]
- If the tangents at the points  $P$  and  $Q$  on the circle  $x^2 + y^2 - 2x + y = 5$  meet at the point  $R\left(\frac{9}{4}, 2\right)$ , then the area of the triangle  $PQR$  is  
[2023 (06 Apr Shift 2)]  
(1)  $\frac{5}{4}$   
(2)  $\frac{13}{8}$   
(3)  $\frac{5}{8}$   
(4)  $\frac{13}{4}$
- Consider a circle  $C_1 : x^2 + y^2 - 4x - 2y = \alpha - 5$ . Let its mirror image in the line  $y = 2x + 1$  be another circle  $C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$ . Let  $r$  be the radius of  $C_2$ . Then  $\alpha + r$  is equal to -----  
[2023 (08 Apr Shift 1)]
- Let  $O$  be the origin and  $OP$  and  $OQ$  be the tangents to the circle  $x^2 + y^2 - 6x + 4y + 8 = 0$  at the points  $P$  and  $Q$  on it. If the circumcircle of the triangle  $OPQ$  passes through the point  $\left(\alpha, \frac{1}{2}\right)$ , then a value of  $\alpha$  is  
[2023 (08 Apr Shift 2)]  
(1)  $\frac{3}{2}$   
(2)  $\frac{1}{2}$   
(3)  $\frac{5}{2}$   
(4) 1
- If the radius of the largest circle with centre  $(2, 0)$  inscribed in the ellipse  $x^2 + 4y^2 = 36$  is  $r$ , then  $12r^2$  is equal to  
[2023 (11 Apr Shift 2)]  
(1) 115  
(2) 92  
(3) 69  
(4) 72
- Two circles in the first quadrant of radii  $r_1$  and  $r_2$  touch the coordinate axes. Each of them cuts off an intercept of 2 units with the line  $x + y = 2$ . Then  $r_1^2 + r_2^2 - r_1r_2$  is equal to \_\_\_\_\_.  
[2023 (12 Apr Shift 1)]
- Let the tangent and normal at the point  $(3\sqrt{3}, 1)$  on the ellipse  $\frac{x^2}{36} + \frac{y^2}{4} = 1$  meet the  $y$ -axis at the points  $A$  and  $B$  respectively. Let the circle  $C$  be drawn taking  $AB$  as a diameter and the line  $x = 2\sqrt{5}$  intersect  $C$  at the points  $P$  and  $Q$ . If the tangents at the points  $P$  and  $Q$  on the circle intersect at the point  $(\alpha, \beta)$ , then  $\alpha^2 - \beta^2$  is equal to  
[2023 (13 Apr Shift 1)]  
(1) 61  
(2) 60  
(3)  $\frac{304}{5}$   
(4)  $\frac{314}{5}$
- Let the centre of a circle  $C$  be  $\alpha, \beta$  and its radius  $r < 8$ . Let  $3x + 4y = 24$  and  $3x - 4y = 32$  be two tangents and  $4x + 3y = 1$  be a normal to  $C$ . Then  $(\alpha - \beta + r)$  is equal to  
[2023 (13 Apr Shift 2)]  
(1) 7  
(2) 5  
(3) 6  
(4) 9
- The number of common tangents, to the circles  $x^2 + y^2 - 18x - 15y + 131 = 0$  and  $x^2 + y^2 - 6x - 6y - 7 = 0$ , is  
[2023 (15 Apr Shift 1)]  
(1) 3  
(2) 1  
(3) 4  
(4) 2

## ANSWER KEYS

1. (121)      2. (3)      3. (2)      4. (3)      5. (2)      6. (7)      7. (3)      8. (1)  
9. (1)

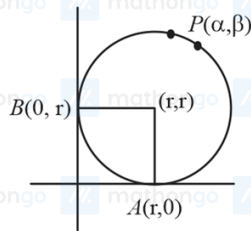
1. (121)

Let the equation of circle touching both the coordinate axis be,

$$(x - r)^2 + (y - r)^2 = r^2 \text{ which passes through } (\alpha, \beta)$$

$$\text{So, } (\alpha - r)^2 + (\beta - r)^2 = r^2 \dots\dots\dots(1)$$

Also equation of line  $AB : x + y = r$



Let  $Q(h, k)$ ,

Now using the formula of foot of perpendicular,

$$\frac{h - \alpha}{1} = \frac{k - \beta}{1} = \frac{-(\alpha + \beta - r)}{2}$$

$$\Rightarrow (h, k) = \left( \alpha - \frac{\alpha + \beta - r}{2}, \beta - \frac{\alpha + \beta - r}{2} \right)$$

$$\Rightarrow (h, k) = \left( \frac{\alpha - \beta + r}{2}, \frac{-\alpha + \beta + r}{2} \right)$$

Now using the distance formula we get,

$$(PQ)^2 = \left( \frac{-\alpha - \beta + r}{2} \right)^2 + \left( \frac{-\alpha - \beta + r}{2} \right)^2 = 121$$

$$\Rightarrow (r - (\alpha + \beta))^2 = 242$$

$$\Rightarrow r^2 + (\alpha + \beta)^2 - 2r(\alpha + \beta) = 242$$

$$\Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta + r^2 - 2r\alpha - 2r\beta = 242$$

$$\Rightarrow 2\alpha\beta = 242 \left\{ \text{from equation (1), } (\alpha - r)^2 + (\beta - r)^2 = r^2 \right\}$$

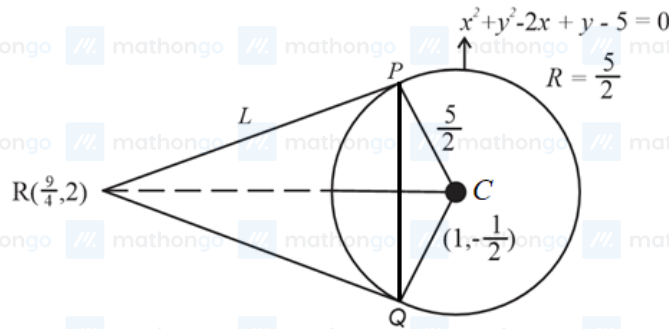
$$\Rightarrow \alpha\beta = 121$$

2. (3)

Given,

The tangents at the points  $P$  and  $Q$  on the circle  $x^2 + y^2 + 2x + y = 5$  meet at the point  $R\left(\frac{9}{4}, 2\right)$ ,

Now plotting the diagram of the above data we have,



Now we know that,

Length of tangent is given by,  $PR = QR = \sqrt{S_1}$

$$\text{So, } L = \sqrt{S_1} = \sqrt{\left(\frac{9}{4}\right)^2 + (2)^2 - 2 \times \frac{9}{4} + 2 - 5}$$

$$\Rightarrow L = \frac{5}{4}$$

$$\text{Now distance between } CR \text{ will be, } CR = \sqrt{\left(\frac{9}{4} - 1\right)^2 + \left(2 + \frac{1}{2}\right)^2} = \sqrt{\frac{25}{16} + \frac{25}{4}} = \sqrt{\frac{125}{16}} = \frac{5\sqrt{5}}{4}$$

Now finding  $PQ$  by equating area, we get

$$\frac{1}{2} \times \frac{5}{2} \times L = \frac{1}{2} \times \frac{PQ}{2} \times CR$$

$$\Rightarrow \frac{1}{2} \times \frac{5}{2} \times \frac{5}{4} = \frac{1}{2} \times \frac{PQ}{2} \times \frac{5\sqrt{5}}{4}$$

$$\Rightarrow PQ = \sqrt{5}$$

Now finding area of triangle  $PQR$  by heron's formula,

$$s = \frac{2\sqrt{5}+5}{4}$$

$$\text{So, area will be } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Rightarrow \Delta = \sqrt{\frac{2\sqrt{5}+5}{4} \left(\frac{2\sqrt{5}+5}{4} - \frac{5}{4}\right) \left(\frac{2\sqrt{5}+5}{4} - \frac{5}{4}\right) \left(\frac{2\sqrt{5}+5}{4} - \sqrt{5}\right)}$$

$$\Rightarrow \Delta = \sqrt{\frac{2\sqrt{5}+5}{4} \left(\frac{2\sqrt{5}}{4}\right) \left(\frac{2\sqrt{5}}{4}\right) \left(\frac{5-2\sqrt{5}}{4}\right)}$$

$$\Rightarrow \Delta = \sqrt{\left(\frac{5}{4}\right) \left(\frac{25-20}{16}\right)} = \frac{5}{8}$$

3. (2)

Given,

$$C_1 : x^2 + y^2 - 4x - 2y + (5 - \alpha) = 0$$

So, its centre will be,  $O_1 = (2, 1)$  and radius  $= \sqrt{\alpha}$

$$\text{And } C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$$

$$\Rightarrow C_2 : x^2 + y^2 - 2fx - 2gy + \frac{36}{5} = 0$$

$$\text{So, Centre } O_2 = (f, g) \text{ and radius } r = \sqrt{f^2 + g^2 - \frac{36}{5}}$$

Also given  $O_2$  is reflection of  $O_1$  in  $2x - y + 1 = 0$ , so image formula we get,

$$\Rightarrow \frac{f-2}{2} = \frac{g-1}{-1} = -2 \cdot \left(\frac{2 \times 2 - 1 + 1}{2^2 + 1^2}\right)$$

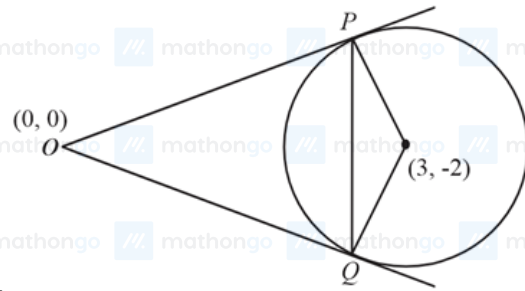
$$\Rightarrow f = \frac{-6}{5} \text{ and } g = \frac{13}{5}$$

$$\text{So, radius } r = \sqrt{\left(\frac{-6}{5}\right)^2 + \left(\frac{13}{5}\right)^2 - \frac{36}{5}} = \frac{\sqrt{25}}{5} = 1$$

$$\Rightarrow r = 1 \text{ and } \alpha = 1 \text{ as they both are same radius circle,}$$

$$\text{Hence, } r + \alpha = 2.$$

4. (3)



The given information can be represented in the form of the diagram below.

Since angle in a semicircle is a right angle,

Hence, the other circle will be passing through the centre of the first circle.

The two ends of diameter of the circle is  $O(0, 0)$  and  $C(3, -2)$ .

Hence, the required equation of circle is  $(x - 0)(x - 3) + (y - 0)(y - (-2)) = 0$

$$\Rightarrow x^2 + y^2 - 3x + 2y = 0$$

Put  $\left(\alpha, \frac{1}{2}\right)$  in the above equation  $\Rightarrow \alpha^2 + \frac{1}{4} - 3\alpha + 1 = 0$

$$\Rightarrow \alpha^2 - 3\alpha + \frac{5}{4} = 0$$

$$\Rightarrow 4\alpha^2 - 12\alpha + 5 = 0$$

$$\Rightarrow 4\alpha^2 - 10\alpha - 2\alpha + 5 = 0$$

$$\Rightarrow 2\alpha(2\alpha - 5) - 1(2\alpha - 5) = 0 \Rightarrow \alpha = \frac{1}{2}, \frac{5}{2}$$

Therefore this is the correct option.

5. (2)

Given equation of ellipse is  $\frac{x^2}{36} + \frac{y^2}{9} = 1$

It is of the form  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

With  $a = 6$ ,  $b = 3$ .

The centre of the circle is  $C(2, 0)$ .

Let  $P(a \cos \theta, b \sin \theta)$  be the common point for ellipse and circle.

$$P \equiv (6 \cos \theta, 3 \sin \theta)$$

Equation of normal to the ellipse will be  $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$ .

$$\Rightarrow 6x \sec \theta - 3y \csc \theta = 36 - 9$$

But the normal through  $P$  passes through the centre of the circle  $C(2, 0)$ .

$$\Rightarrow 6(2) \sec \theta - 3(0) \csc \theta = 27$$

$$\Rightarrow \sec \theta = \frac{27}{12}$$

$$\Rightarrow \cos \theta = \frac{4}{9} \text{ and } \sin \theta = \frac{\sqrt{65}}{9}$$

$$P \equiv \left(6 \times \frac{4}{9}, 3 \times \frac{\sqrt{65}}{9}\right)$$

$$P \equiv \left(\frac{8}{3}, \frac{\sqrt{65}}{3}\right)$$

$$\text{Now } CP = r = \sqrt{\left(2 - \frac{8}{3}\right)^2 + \left(0 - \frac{\sqrt{65}}{3}\right)^2}$$

$$\Rightarrow r^2 = \frac{69}{9}$$

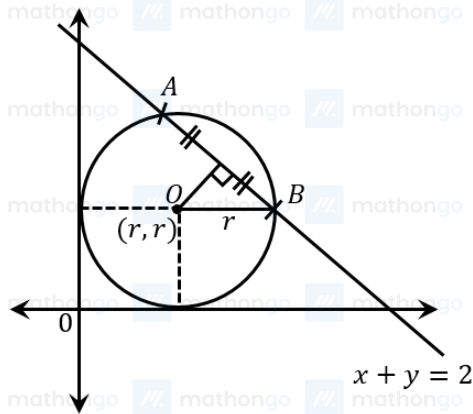
$$\Rightarrow 12r^2 = 12 \times \frac{69}{9} = 92$$

Therefore, the required answer is 92

6. (7)

Assuming a circle of radius  $r$  which touches both the axis is given by,  $(x - r)^2 + (y - r)^2 = r^2$

Now given line  $x + y = 2$  makes the intercept 2 the circle, so plotting the diagram we get,



Now from diagram,

$$AC = BC = 1, OC = \sqrt{r^2 - 1}$$

Now using the formula of perpendicular distance of point from the line we get,

$$OC = \left| \frac{2r - 2}{\sqrt{2}} \right| = \sqrt{r^2 - 1}$$

$$\Rightarrow 2(r - 1)^2 = r^2 - 1$$

$$\Rightarrow r^2 - 4r + 3 = 0$$

$$\Rightarrow r_1 = 1, r_2 = 3$$

$$\text{Hence, } r_1^2 + r_2^2 - r_1 r_2 = 1 + 9 - 3 = 7$$

7. (3)

Given that  $\frac{x^2}{36} + \frac{y^2}{4} = 1$

Observe that  $(3\sqrt{3}, 1)$  lies on the given ellipse.

Equation of tangent to the ellipse at the given point is  $T : \frac{3\sqrt{3}x}{36} + \frac{y}{4} = 1$

$$\Rightarrow \frac{\sqrt{3}x}{12} + \frac{y}{4} = 1$$

Equation of normal passing through same point will be  $N : \frac{x-3\sqrt{3}}{\frac{3\sqrt{3}}{36}} = \frac{y-1}{\frac{1}{4}}$

$$\Rightarrow \frac{12x-36\sqrt{3}}{\sqrt{3}} = 4y-4$$

$$\Rightarrow 3x-9\sqrt{3} = \sqrt{3}y - \sqrt{3}$$

$$\Rightarrow 3x - \sqrt{3}y = 8\sqrt{3}$$

Now tangent and normal intersects the  $y$ -axis at  $A$  and  $B$  respectively.

$$A(0, 4) \quad B(0, -8)$$

Equation of circle with  $AB$  as diameter will be

$$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$$

$$\Rightarrow x^2 + (y-4)(y+8) = 0$$

Line  $x = 2\sqrt{5}$  intersects the circle.

$$\Rightarrow 20 + y^2 + 4y - 32 = 0$$

$$\Rightarrow y^2 + 4y - 12 = 0$$

$$\Rightarrow (y+6)(y-2) = 0$$

$$\text{Hence } P(2\sqrt{5}, -6) \quad Q(2\sqrt{5}, 2)$$

Now let us find the equation of tangents to the circle at  $P, Q$ .

$$\Rightarrow T : xx_1 + yy_1 + 2y + 2y_1 - 32 = 0$$

$$\Rightarrow T_1 : 2\sqrt{5}x - 6y + 2y - 12 - 32 = 0$$

$$\Rightarrow 2\sqrt{5}x - 4y = 44$$

$$\Rightarrow T_1 : \sqrt{5}x - 2y = 22 \dots \dots \dots (i)$$

$$\Rightarrow T_2 : 2\sqrt{5}x + 2y + 2y + 4 - 32 = 0$$

$$\Rightarrow 2\sqrt{5}x + 4y = 28$$

$$\Rightarrow T_2 : \sqrt{5}x + 2y = 14 \dots \dots \dots (ii)$$

From (i) and (ii)

$$\alpha = \frac{18}{\sqrt{5}} \quad \beta = -2$$

$$\Rightarrow \alpha^2 - \beta^2 = \frac{304}{5}$$

Hence this is the required option.

8. (1)

Given,

The centre of a circle  $C$  be  $\alpha, \beta$  and its radius  $r < 8$ ,

And  $3x + 4y = 24$  and  $3x - 4y = 32$  be two tangents

So, the distance from centre will be radius,

$$\left| \frac{3\alpha + 4\beta - 24}{5} \right| = \left| \frac{3\alpha - 4\beta - 32}{5} \right|$$

Taking + sign we get,

$$\Rightarrow 3\alpha + 4\beta - 24 = 3\alpha - 4\beta - 32$$

$$\Rightarrow 8\beta = -8 \Rightarrow \beta = -1$$

And taking - sign we get,

$$3\alpha + 4\beta - 24 = -3\alpha + 4\beta + 32$$

$$\Rightarrow 6\alpha = 56 \Rightarrow \alpha = \frac{56}{6}$$

And  $4x + 3y = 1$  be a normal to  $C$

So,  $(\alpha, \beta)$  lies on  $4x + 3y = 1$

$$\Rightarrow 4\alpha + 3\beta = 1$$

$$\text{So, } \alpha = 1 \text{ when } \beta = -1 \text{ and } \beta = \frac{1}{3} \left( 1 - 4 \times \frac{28}{3} \right) = \frac{-109}{9}, \text{ if } \alpha = \frac{28}{3}$$

Hence, radius will be,

$$r = \left| \frac{3 - 4 - 24}{5} \right| = 5 < 8 \text{ if } (\alpha, \beta) = (1, -1)$$

And if  $(\alpha, \beta) = \left( \frac{28}{3}, \frac{-109}{9} \right)$  then  $r > 8$ , so neglected.

$$\therefore r = 5, \alpha = 1, \beta = -1$$

$$\therefore \alpha - \beta + r = 7$$

9. (1)

$$\text{If } S \equiv x^2 + y^2 + 2gx + 2fy + c = 0.$$

$$C \equiv (-g, -f) \text{ and } r = \sqrt{g^2 + f^2 - c}.$$

The given circles are  $x^2 + y^2 - 18x - 15y + 131 = 0$

$$\Rightarrow C_1 \left( 9, \frac{15}{2} \right), r_1 = \sqrt{81 + \frac{225}{4} - 131} = \frac{5}{2}$$

$$\text{and } x^2 + y^2 - 6x - 6y - 7 = 0$$

$$\Rightarrow C_2(3, 3), r_2 = \sqrt{9 + 9 - 7} = 5$$

$$\Rightarrow d = C_1 C_2 = \sqrt{(9 - 3)^2 + \left( \frac{15}{2} - 3 \right)^2} = \sqrt{36 + \frac{81}{4}} = \frac{15}{2}$$

$$\Rightarrow r_1 + r_2 = \frac{5}{2} + 5 = \frac{15}{2}$$

$$\Rightarrow C_1 C_2 = r_1 + r_2$$

$\therefore$  Circles touch each other externally, 3 common tangents.

Hence this is the correct option.